Data Cleaning and Preprocessing Project

Overview

This project focuses on cleaning and preprocessing a customer dataset using Python and Pandas. The goal is to transform raw data into a clean, structured, and analysis-ready format.

Objectives

- Load and inspect the dataset.
- Identify and remove duplicate records.
- Check and handle missing values.
- Standardize text fields for consistency.
- Convert data types where necessary.
- Validate the cleaned dataset.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Google Colab

Data Cleaning Steps

1. Data Loading

The customer dataset was imported using the Pandas library.

2. Duplicate Removal

Duplicate records were identified and removed to improve data quality.

3. Missing Value Analysis

Missing values were checked across all columns and handled appropriately.

4. Data Standardization

- Converted text fields to a consistent format.
- Removed leading and trailing whitespaces.
- Standardized email, city, and country values.

5. Data Type Conversion

Date columns were converted into the appropriate datetime format for better analysis.

6. Validation

The dataset was validated to ensure:

- No duplicate records remained.
- Missing values were handled correctly.
- Data formatting was consistent.

Output

The cleaned dataset was exported and saved as:

`cleaned_customers.csv`

Results

The preprocessing pipeline successfully transformed the raw customer dataset into a clean, structured, and analysis-ready dataset suitable for further analysis and visualization.

Author

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